

GB/s; and Gen 5 lifts it to over 14 GB/s. More bandwidth means faster load times and transfers, but for everyday web browsing or office work you won't notice the difference between Gen 4 and Gen 5. Most gamers and creators find Gen 4 SSDs strike the right balance of speed and price. High-end Gen 5 drives push performance but consume more power and run hot, requiring proper cooling. Mainstream Gen 5 models compromise slightly on speed to keep temperatures manageable.

Gen 5 NVMe drives are backward-compatible with Gen 4 slots, albeit at Gen 4 speeds. That makes them a decent purchase if you plan to upgrade the motherboard later. Always check that your board has enough PCIe lanes and physical M.2 slots for the drive you want.

NAND types

Solid-state drives store data in NAND flash cells. Triple-level cell (TLC) NAND holds three bits per cell (eight voltage states), while quad-level cell (QLC) NAND packs four bits (16 states). QLC's higher density yields lower cost per gigabyte but slows read and write operations and reduces endurance. TLC writes faster and lasts longer, so it's used in performance-oriented consumer and professional drives.

QLC suits read-heavy workloads such as media libraries or AI inference where capacity matters more than write speed. Choose TLC if you install games, run virtual machines or generate lots of random writes. A few enterprise drives use pseudo-SLC modes that force cells to store only one bit, trading capacity for vastly increased endurance; the concept illustrates how more voltage states reduce longevity.



Endurance and workload patterns

Manufacturers quote endurance using total bytes written (TBW) or drive writes per day (DWPD). TBW gives the cumulative amount of data you can write before the drive might fail; DWPD indicates how many full-drive writes you can perform daily during the warranty period. These figures are guidelines: real-world endurance varies with workload type, temperature and power cycles. Sequential writes (copying large files) are gentler on NAND than random updates because the controller writes contiguous blocks. Drives may publish different ratings for sequential and random workloads. Another factor is over-provisioning – manufacturers reserve extra flash to spread wear across more cells. Some vendor utilities let you reduce usable capacity to extend life. If you plan to archive data, remember that unpowered SSDs slowly lose charge; cooler storage prolongs data retention. When buying, look at the TBW rating and consider whether your workload is mostly sequential (video editing, backups) or random (operating systems, databases).

TLC VS QLC NAND: UNDERSTANDING ENDURANCE AND VALUE

Attribute	TLC (Triple-Level Cell)	QLC (Quad-Level Cell)	What You Should Know
Bits Stored per Cell	3 bits	4 bits	QLC stores more data per cell but is less durable.
Speed Profile	Faster read/write	Slower writes without cache	TLC handles frequent small writes better.
Endurance (TBW Ratings)	Higher (> 600 TBW for 1 TB drive)	Lower (~300 TBW for 1 TB drive)	TLC lasts roughly 2× longer.
Cache Mechanism	Static + dynamic SLC cache	Heavily relies on SLC cache	QLC speed collapses when cache fills.
Ideal Use Case	Gaming, creative, everyday OS drive	Large media storage, backups	QLC suits read-heavy, infrequent-write workloads.
Power Efficiency	Moderate	Slightly better idle efficiency	Battery-saver edge in portable setups.
Cost per GB	Higher (~₹8 – ₹10 / GB)	Lower (~₹5 – ₹7 / GB)	QLC offers more capacity for the price.
Example Products	Samsung 990 Pro (TLC), Crucial P5 Plus (TLC)	WD Blue SN580 (QLC), ADATA XPG Spectrix S20 (QLC)	Illustrative brand references only.